Carryover-Representation and Half-Life-Selection Methods for Paper 02: A Catalog of the G- and F-Families

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Purpose. Two separate, only partly reconciled lines of work bear on which analysis specification paper 02 should recommend. The first is paper 02's own nine specifications, `G1-G9`, compared on a simple model (`Sx ~ bm + t + X + bm:X`, no expectancy term) that differ only in how carryover enters as `X`. The second is a project-wide research thread on choosing the analysis-side carryover half-life, built on the richer "F1" model (`Y ~ bmc + t + Dbc + bmc:Dbc + De + bmc:De`, with expectancy terms), documented in `docs/57-data-chosen-carryover-half-life.md` and `docs/series/ch23-data-chosen-half-life.md`. This note catalogs both families in one place: what each method is, what has been determined about it so far (evidence-labeled), and its approximate computational cost where available. It does not propose a recommendation; it is a reference document for deciding what paper 02's own remediation (`docs/60-trials-submission-gap-analysis.md`, Part 2) should test next.

What "best" would mean. No method cataloged here has been checked against all of the following, and that gap is itself the main finding of this note. Reading the "Determined" entries against this list is the intended way to use the catalog — a method's entry states which of these it has been shown to satisfy, which it has been shown to fail, and which remain untested, rather than offering a single verdict.

1. **Type I error (size).** Does the test hold its nominal level under the null, across the cells actually checked — not just at the one reference cell most specifications have been tested at?
2. **Power**, both in absolute terms and relative to the other candidates, at the reference cell and, ideally, across the grid.
3. **Robustness to carryover half-life misspecification** — does performance hold when the analyst's assumed half-life (or the method's own data-driven estimate of it) diverges from the true one?
4. **Robustness to the decay-shape assumption** (exponential vs. Weibull, light- vs. heavy-tailed) — does the method depend on getting the *shape* of carryover decay right, not just its rate?
5. **Robustness to trial design** — does a specification validated at the Hybrid reference cell still perform well under CO or OL+BDC, or is its apparent advantage a Hybrid-specific artifact (as `G3`/`G4` turned out to be)?
6. **Robustness to the DGP architecture** (covariance vs. mean moderation) — does the ranking among specifications survive a change in how the biomarker-treatment interaction is encoded?
7. **Robustness to a prognostic biomarker** — does the method still hold nominal size when the biomarker is also associated with baseline severity or the natural course, not only predictive of treatment response? Paper 01's Study 3 found this the single sharpest discriminator among otherwise-similar analyses, and no `G`-family specification has been tested against it at all.
8. **Computational cost**, both per-replicate (for simulation studies) and per-trial (for analyzing a single real dataset), since methods that scale by a bootstrap multiplier M are impractical for some uses and routine for others.
9. **Validation completeness itself** — how many of 1-8 have actually been checked for a given method, as distinct from how well it did on the ones that have been. `G8` and `F1` are furthest along this axis; `G9` is the best performer on power alone but the least validated against the rest of this list.

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Notation and evidence labels

`Db`: binary on-drug indicator. `Dbc`: continuous exposure-decayed drug indicator, `1` on drug, decaying with time since discontinuation off drug, `0` before first exposure. `L`: lagged-treatment nuisance indicator (Jones and Kenward). `bm`/`bmc`: biomarker, raw or centered. `De`: design-expectancy covariate (η_t); absent from the `G`-family, present in the `F`-family. $t^a_{1/2}$: the analysis-side carryover half-life used to build `Dbc`. CR2: Bell-McCaffrey bias-reduced cluster-robust standard error, with Satterthwaite degrees of freedom. $\mathcal{T}_a$: the candidate half-life grid. Paper 02 uses $\mathcal{T}_a = \{0.25, 0.5, 1.0, 2.0\}$ weeks; `docs/57/ch23` use $\mathcal{T}_a = \{0, 0.25, 0.5, 1, 2\}$ weeks (one more point, including 0).

Each claim below carries one of four evidence labels, the project convention: *quoted* (text reproduced from a source), *inspected* (read from source code or a manuscript table), *derived* (follows algebraically from a quoted or inspected fact) or *verified* (computed by running code in this or an earlier session). Status entries not otherwise labeled are *inspected* summaries of the cited section's reported numbers.

Part 1: the G-family (paper 02's own specifications)

All nine share the base model `Sx ~ bm + t + X + bm:X`, fit by `nlme::lme` with a random intercept and continuous-time AR(1) residuals (`corCAR1`); none include an expectancy term. Source: `analysis/report/02-carryover-sensitivity/report.Rmd`, Table `tab-spec-summary` and the sections cited per row below.

Table 1: The `G`-family, quick index

Code	Name	Formula for X
`G1`	Unadjusted (`E1`)	$X = D_{it}$, binary
`G2`	Lag-adjusted (`E3`)	$X = D_{it}$ + nuisance $L_{it} = D_{i,t-1}(1 - D_{it})$
`G3`	Exposure-weighted (`E2`)	$X = D_{bc,it}$, half-life assumed/fixed
`G4`	AIC-selected $t^a_{1/2}$ (`E7`)	$X = D_{bc,it}$, $t^a_{1/2}$ by AIC, interaction already in the selection model
`G5`	Paired-difference (`E9`)	per-patient on/off mean difference, OLS on biomarker
`G6-G8`	`G1`/`G2`/`G3` + CR2	same X as base spec
`G9`	`G4` + CR2 (`E7cr2`)	same X as `G4`
retired	`E4`/`E5`/`E6`	biomarker-interacted carryover terms

`G1` Unadjusted (`E1`)

Determined. Baseline benchmark; no carryover representation. Statistically indistinguishable from `G2` everywhere tested (largest gap 0.003-0.004 across every Tier 1 and Tier 2 cell; identical to two decimal places in the higher-precision 24-cell paired rerun) — the manuscript's clearest negative result. Trails `G3-G5` at the Hybrid/OL+BDC reference cell (0.760 vs. 0.836-0.850 at $t_{1/2} = 1.0$) but is the best-performing specification under CO (0.834, vs. `G3`/`G4` collapsing to 0.48-0.55). CR2 (`G6`) lifts it modestly (0.748 → 0.776) without changing its ranking relative to `G3-G5`.

Cost. Cheapest of the nine: one `lme` fit per replicate.

`G2` Lag-adjusted (`E3`)

Determined. Adds the classical Jones-and-Kenward lagged-treatment term. Does no measurable work anywhere in the grid: indistinguishable from `G1` at every half-life, design, architecture and Tier 2 block checked. CR2 (`G7`) gives the same modest, ranking-preserving gain as `G1`→`G6` (0.748 → 0.764).

Cost. Same order as `G1`: one extra fixed-effect column, negligible extra compute.

`G3` Exposure-weighted (`E2`)

Determined. The current validated default and the most thoroughly tested of the nine. Beats `G1`/`G2` at Hybrid/OL+BDC; holds under half-life mis-specification short of a large over-assumption (Block S2) and under light-tailed decay-shape mis-specification (narrowing to non-significance only at the two most extreme heavy-tailed shapes, Section 3.5). CR2 (`G8`) restores nominal size without disturbing the ranking (Block S6; validated at $N \in \{35, 70\}$ in Hybrid specifically, not itself stress-tested in OL+BDC). Fails markedly under CO (0.482-0.546 vs. `G1`/`G2`'s 0.834-0.866) — this is inherited by `G4` almost entirely (Section 3.9). Reverses marginally under mean moderation (Block S8: trails `G1`/`G2` by 2-3 points instead of leading).

Cost. One `lme` fit at a single pre-specified half-life — same order as `G1`/`G2`, no grid search.

`G4` AIC-selected $t^a_{1/2}$ (`E7`)

Selects $t^a_{1/2} = \arg\min_{\mathcal{T}_a}$ AIC of the model **with** `bm:Dbc` already in it.

Determined. Confirmed 2026-10-21: this is the liberal "F1A"-style selection (same pattern as the package's own `characterize_carryover()`, `R/carryover_analysis.R`). Beats `G3` at the Hybrid/OL+BDC reference cell (0.850 vs. 0.836 at $t_{1/2} = 1.0$). Fails catastrophically under CO: AIC drifts to the longest grid candidate in 86-97% of replicates regardless of the true half-life, but this mis-selection costs almost nothing beyond `G3`'s own CO collapse with the true half-life in hand (0.474 vs. 0.482) — the dominant failure is Exposure-weighted's CO weakness, not the AIC step itself. Type I error (Block S11, one cell, $n = 2000$): 0.038, "not inflated" — but this session's quick replication (500 reps, corrected DGP) reproduced 0.038 at the same cell, while `docs/57`/`ch23`'s pooled check of the same naive-selection pattern over 11 null cells found 0.067-0.074; whether `G4`'s single tested cell represents the grid is open. Not yet tested with an expectancy term or a prognostic biomarker (paper 01 Study 3 found exactly that combination breaks simpler models).

Cost. Five `lme` ML fits (one per $\mathcal{T}_a$ candidate) plus one REML refit per replicate — roughly 5-6× a single `G3` fit. Matches `docs/57` Method 2's reported cost for the same fit count (~0.5 sec/trial, *quoted*, Section 4.2). This session's quick check: ~0.6 sec/replicate running an `E7`-equivalent alongside an `F1A0`-style comparator together (500 reps, ~303 sec total).

`G5` Paired-difference (`E9`)

Per-patient on-drug minus off-drug mean, OLS on biomarker; no carryover term, no repeated-measures structure.

Determined. Beats `G1` at the reference cell (0.842) but trails `G4` narrowly. The most architecture-robust of the five Tier 1 candidates — never reverses to a large disadvantage either way (Block S8: 0.964 under mean moderation vs. `G1`/`G2`'s 0.972/0.976, a smaller gap than `G3`/`G4`'s). Underperforms `G1`/`G2` under CO but mildly (0.750-0.778 vs. 0.834-0.866), reflecting that CO's own within-subject contrast already captures most of the signal a paired-difference regression targets. No CR2 variant applies (collapses to one row per patient before fitting). Tested at only one design/cell pairing outside the Hybrid comparison (Block S9, CO only).

Cost. Cheapest of the data-driven alternatives: a single OLS fit after a deterministic per-patient aggregation — no `nlme::lme` call at all, materially faster than `G3`/`G4`.

G6-G8 (G1/G2/G3 + CR2)

Determined. CR2 lifts every base specification 2-5 points without disturbing the ranking among G1-G5 (Block S6). G8 (G3+CR2) is the validated extension of the current default, checked against the same robustness battery as G3 itself.

Cost. One extra clubSandwich/Satterthwaite computation on top of the base fit; small fixed overhead.

G9 (E7cr2, G4+CR2)

Determined. The single best power result anywhere in the manuscript (0.870 at the reference cell, a 3.8-point CR2 gain over G4's 0.832 — the largest CR2 correction for any specification), overtaking G8 (0.863). Also the least validated: rests on one cell, with none of G4's own robustness gaps (CO failure, mean-moderation reversal, untested prognostic-biomarker behavior) closed, and its Type I error claim (0.052 at that one cell) carries the same single-cell-generalization caveat as G4's.

Cost. Same grid-search cost as G4, plus CR2's small fixed overhead.

Retired E4/E5/E6 (biomarker-interacted carryover terms)

A lagged term, and a continuous elapsed-time term, each crossed with bm.

Determined. Underperformed G1 at every cell examined across three blocks, two designs, two architectures and three half-lives. Traced to a structural cause: the DGP has a single true decay parameter, so crossing a second carryover term with the biomarker adds a free parameter with no counterpart in the truth being estimated. Replaced by G4 and G5.

Cost. Not reported separately; superseded before production-scale timing was recorded.

Batch-level run times (quoted from report.Rmd's Reproducibility section; 8 cores unless noted), for context on the relative cost of running the full nine-specification grid versus a subset:

Table 2: Batch run times by block, as reported in report.Rmd's Reproducibility section (*quoted*)

Run	Cells	Specs	n_{sim}	Time
Tier 1 grid, G1-G9 extension	30	all 9	500	~3300 sec (~55 min)
Decay-shape sensitivity, G1-G3 only	30	3	500	~770 sec
Decay-shape sensitivity, G1-G9 extension	30	all 9	500	~2500 sec
S1-S4 G1-G9 extension	25	all 9	500	~13800 sec (well above linear scaling from the 100-rep run; not further diagnosed, *quoted*)
S6 G1-G9 extension	10	all 9	500	~1210 sec
S7	1	3	500	~40 sec
S8 (architecture sensitivity)	2 architectures × 3 half-lives	5	500 each	~5.5 min
S9 (CO)	2	5 (G1-G5)	500	~8 min

Run	Cells	Specs	n_{sim}	Time
S10 (CR2 comparison)	1	8	500	~2 min
S11 (Type I error)	1	6 (`G1-G5,G9`)	2000	~7 min
S12 (mean-moderation counterpart, Tier 1)	30	all 9	500	~55 min on 8 cores

Part 2: the F-family (half-life-selection methods, `docs/57/ch23`)

All built on the richer model `Y ~ bmc + t + Dbc + bmc:Dbc + De + bmc:De`; the methods differ only in how $t^a_{1/2}$ (feeding `Dbc`) is chosen. Source: `docs/57-data-chosen-carryover-half-life.md`, section numbers as cited.

Table 3: The `F`-family, quick index

Method	What it does
F1	$t^a_{1/2}$ fixed in advance (0.2 week)
F1A	AIC over $\mathcal{T}_a$, interaction already in the selection model
F1A0 (+ F1G)	AIC over $\mathcal{T}_a$ on the null model; F1G is a guarded fixed variant
Method 3 (DS/DA/ES/EA)	sample-splitting and cross-fitting
Method 4	profile-likelihood interval for $t^a_{1/2}$, not a test
Method 6	joint/continuous estimation of $t^a_{1/2}$
F1AB / F1A0B	bootstrap calibration of F1A / F1A0
Method 7 (MaxT)	$\max_h \lvert T(h)\rvert$ over the grid
Method 8 (MA0/MA1/MAB/MA0B)	marginalize over $t^a_{1/2}$ (Bayesian / AIC-weight)
Permutation	residual permutation, Freedman-Lane style

F1 (Method 1, fixed)

$t^a_{1/2}$ fixed in advance (0.2 week, current policy), with a non-inferential sensitivity grid reported alongside.

Determined. Size 0.051 pooled (at most 0.062 in one cell), stored setting; 0.053 pooled over 12 cells under the refined-process extension; at the current 0.2-week policy value, size 0.050 with no power gain over the historical 0.5-week default. "The validated simple fallback." A fixed value can lose real power when far from the best-fitting one (size-adjusted power 0.496 at true half-life 0 vs. 0.272 at true half-life 1 week, fixed at 0.5, *quoted* `docs/47` Section 6).

Cost. One fit per trial; five more for the (non-inferential) sensitivity grid.

F1A (naive)

Narrative baseline, not separately numbered in `docs/57` — the failure case motivating Methods 2 and 5. AIC over $\mathcal{T}_a$, selection model **already includes** the interaction; test as if $t^a_{1/2}$ were fixed.

Determined. Liberal: the same data drive selection and test. Pooled Type I error 0.074 across 11 non-prognostic null cells (vs. F1's 0.061 on the same trials); 0.079 specifically where it selected the longest candidate (2 weeks). This is the pattern `G4`/`G9` currently implement.

Cost. Same fit count as F1A0: five ML fits across the grid plus one REML refit.

F1A0 (Method 2) + F1G guard variant

AIC over $\mathcal{T}_a$ on the model **without** the interaction, refit with it at the selected value; F1G keeps $t^a_{1/2} = 0.5$ fixed unless a null-model profile-likelihood test rejects it.

Determined. Removes "about half" of F1A's excess size, not all of it: 0.067 pooled (illustration) vs. F1A's 0.074; 0.053 pooled over 12 cells in the refined-process extension, matching F1 almost exactly — "as nominal tests neither gains power over F1." The orthogonality argument motivating it is not borne out cleanly at $N = 70$. F1G performs almost identically (0.053 pooled), switching away from the fixed value in only a minority of trials. With bootstrap calibration (→ F1A0B), "the primary data-driven candidate."

Cost. Five ML fits and one REML fit with CR2 per trial, about 0.5 seconds (*quoted*).

Method 3: sample-splitting / cross-fitting (`ch23`'s DS/DA/ES/EA)

Single split: select on half, test on the other half. Cross-fit average: average the two swapped single-split estimates. Bonferroni and Cauchy p-value combination across the split.

Determined. Single split: 0.044 pooled null, power 0.208 (conservative; loses power testing on $N/2$). Cross-fit average: 0.061 pooled, power 0.372 (closer to nominal, more power, but the two halves' statistics are not independent, so this calibration is approximate). Cauchy combination: 0.050 and 0.292 (closest to nominal of the three). Bonferroni is exact and conservative by construction.

Cost. About the cost of F1A0 (ten ML fits on half the data plus two REML fits with CR2); the cost that matters is the power lost testing on half the sample (noncentrality falls by about $\sqrt{2}$ for a single split).

Method 4: report the half-life, not a test

A profile-likelihood interval for $t^a_{1/2}$, reported alongside whichever test is used for inference.

Determined. Not itself a test of the interaction. Illustrated on one trial; coverage not evaluated. "Recommended for reporting in every case" regardless of which test is used.

Cost. 11 seconds per trial (52-point grid, two ML fits per point, plus a REML refit and CR2 for the profile) — once per trial, not per bootstrap resample.

Method 6: continuous/joint estimation

Estimate $t^a_{1/2}$ as a continuous nonlinear parameter rather than selecting from a candidate grid.

Determined. Proposed, not adopted as a working method here. Its main contribution is a theoretical orthogonality argument supporting F1A0's validity case. No CR2 analog exists yet for this nonlinear parameterization.

Cost. One `nlme` fit per trial, typically several times an `lme` fit's cost, with convergence failures reported near parameter boundaries; or a ~25-point profile plus a sandwich correction.

F1AB / F1A0B (Method 5, parametric bootstrap calibration)

Calibrate the select-then-test statistic's critical value via a parametric null-model bootstrap that repeats the entire selection step in every resample.

Determined. This is what actually fixes the size problem. Illustrated (warp-speed bootstrap): F1AB 0.062 pooled (vs. F1A's own 0.061 on the same trials), power 0.364. Evaluated in the refined-process extension: F1A0B 0.050 and F1AB 0.050 pooled, mean size-adjusted power 0.395 and 0.400, maximin shortfall 0.076 and 0.078 — both hold size at this scale. **F1A0B is the project's current primary data-driven**

candidate, still described as "provisional"; it has not been tested in paper 02 or cited by it (see `docs/60-trials-submission-gap-analysis.md`, Part 2, items 7 and 12).

Cost. Each resample costs one run of the selection procedure (~1.1 sec for the five-fit grid). For one trial, $M = 199$ resamples took 128 sec on 7 workers; $M = 999$ would take ~11 minutes. In a simulation study this multiplies per-replicate cost by M unless the "warp-speed" shortcut (one resample per Monte Carlo replicate, pooled across the cell) is used, bringing it to about twice the cost of the plain, uncalibrated procedure.

Method 7: maximum statistic over the grid (MaxT)

Test $\max_{h \in \mathcal{T}_a} |T(h)|$ against its joint null distribution, rather than picking one h; Bonferroni / multivariate-normal / bootstrap-calibrated variants.

Determined. Bonferroni: exact and conservative (0.024 pooled in the illustration; 0.018, "too conservative," in the extension). Multivariate-normal reference: approximate, liberal in the extension (0.072). Warp-speed bootstrap calibration: 0.048 pooled, power 0.384 in the illustration — similar performance to F1A0B. Reports a p-value and the half-life of largest $|T|$, not a single effect-size estimand, which "weakens its face validity for a reader who wants an effect size."

Cost. Five REML fits with CR2 (about 1 second); bootstrap calibration adds M times that per trial, the same multiplier structure as F1AB/F1A0B.

Method 8: marginalizing over the half-life

Average over $\mathcal{T}_a$ weighted by (posterior or AIC-derived) model weight, rather than selecting a single h (Bayesian model averaging; AIC-weight stand-in).

Determined. Full Bayesian MCMC version proposed but not run (cost prohibitive at simulation scale). AIC-weight stand-in illustrated: 0.053 pooled with interaction-included weights, 0.049 with null-model weights; evaluated in the extension: 0.0455 and 0.040 (both slightly conservative); with bootstrap calibration (MAB/MA0B): 0.048 and 0.049, mean size-adjusted power 0.401 and 0.399 — essentially matching F1AB/F1A0B.

Cost. Full Bayesian: an assumed (not measured) 1 to 5 minutes per trial with a custom Stan program, making large-scale calibration expensive. AIC-weight stand-in: no extra fitting cost beyond the grid fits already computed for F1A/F1A0.

Permutation tests

Residual permutation (Freedman-Lane style, within path), as a nonparametric alternative to the bootstrap.

Determined. Naive permutation of the biomarker is invalid when it is prognostic (destroys an association in the resamples that remains in the observed data). A valid within-path residual-permutation version is "proposed with conditions; not recommended over the parametric bootstrap, which needs fewer exchangeability assumptions and handles unequal path sizes and dropout directly."

Cost. Approximately that of the parametric bootstrap, per resample, without a clear cost advantage to offset its added assumptions.

Part 3: how the two families connect

`F1`'s `Dbc` term is the identical construct to `G3`/`G8`'s Exposure-weighted predictor (both: 1 on drug, exponential decay off drug after first exposure, 0 before first exposure). `F1A`/`F1A0`/ `F1AB`/`F1A0B` are therefore asking the same question `G4`/`G9` ask — how to choose $t^a_{1/2}$ — on top of the richer, expectancy-term model, and (for the

B-suffixed variants) with the bootstrap calibration G4/G9 currently lack. Three gaps follow directly from the two tables above:

1. **G4/G9 implement F1A, not F1A0 or F1A0B.** The naive, interaction-included AIC selection is the variant docs/57/ch23 found most liberal. Paper 02 has not tested the cheaper partial fix (F1A0) or the calibrated fix (F1A0B) on its own grid.
2. **The G-family has never been tested with an expectancy term or a prognostic biomarker**; the F-family's methods are all built on a model that already includes one, because paper 01's Study 3 found it necessary. G4/G9's single-cell Type I error check (Section S11) cannot speak to this, since its model lacks the term that was found to matter.
3. **No single document currently compares the two families directly** — G4/G9's power against F1's, or either against F1A0B, on a shared grid. docs/60-trials-submission-gap-analysis.md Part 2, items 11-13 track closing this gap.

References

- analysis/report/02-carryover-sensitivity/report.Rmd, Table tab-spec-summary and Sections 2.4, 2.7, 3.4-3.9.
- docs/57-data-chosen-carryover-half-life.md, Section 4 (Methods 1-9) and Section 8 (the refined-process extension).
- docs/series/ch23-data-chosen-half-life.md, construct box and candidate grid.
- docs/60-trials-submission-gap-analysis.md, Part 2, for the remediation items this catalog supports.
- CLAUDE.md (pmsimstats-ng project file), "Recommended analysis for the interaction (F1)" and "Known open item" entries.